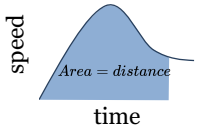
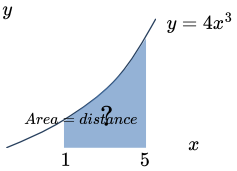
1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 19 · Definite Integration





So far we've seen integration as 'the opposite of differentiation', allowing us to find $y = f(x)$ when we know the gradient function $y = f'(x)$.

In practical settings however the most useful use of integration is that it finds the area under a graph. Remember at GCSE for example when you estimated the area under a speed-time graph, using trapeziums, to get the distance?

If you knew the equation of the curve, you could get the exact area!

Before we do this, we need to understand how to find a definite integral:

These are known as limits, which give the values of x we're finding the area between.

We integrate as normal, but put expression in square brackets, meaning we still need to evaluate the integrated expression using the limits.

$\int_1^5 4x^3 dx =$

Write $(\dots) - (\dots)$ and evaluate the expression for each of the limits, top one first.

Chapter 13 · Original slide 20 · Another Example

We DON'T have a constant of integration when doing definite integration. I'll explain why later.

Write out you working EXACTLY as seen here. The $(\dots) - (\dots)$ brackets are particularly crucial as you'll otherwise likely make a sign error.

You can use the $\left[\left[\int_b^a \square\right]\right]$ button on your calculator to evaluate definite integrals.

But only use it to check your answer.

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 21 · Problem Solving

Given that P is a constant and $\int_1^5 (2Px + 7)dx = 4P^2$, show that there are two possible values for P and find these values.

Remember: P is a constant, so just treat it as a number.

Chapter 13 · Original slide 22 · Exercise 13D – Y1 page 297

4 Given that A is a constant and $\int_1^4 (6\sqrt{x} - A)\,\mathrm{d}x = A^2$, show that there are two possible values for

A and find these values.

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 23 · Exercise 13D – Y1 page 297

5 Use calculus to find the value of $\int_1^9 (2x - 3\sqrt{x})\,\mathrm{d}x$.

6 Evaluate $\int_4^{12} \frac{2}{\sqrt{x}}\,\mathrm{d}x$, giving your answer in the form $a + b\sqrt{3}$, where a and b are integers.

Chapter 13 · Original slide 24 · Exercise 13D – Y1 page 297

7 Given that $\int_1^k \frac{1}{\sqrt{x}}\,\mathrm{d}x = 3$, calculate the value of k.

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 25 · EXTENSION

2

[MAT 2015 1D] Let

$f(x) = \int_0^1 (xt)^2\,dt$ and $g(x) = \int_0^x t^2\,dt$

Let $A > 0$. Which of the following statements are true?

$g(f(A))$ is always bigger than $f(g(A))$

$f(g(A))$ is always bigger than $g(f(A))$

They are always equal.

$f(g(A))$ is bigger if $A < 1$, and $g(f(A))$ is bigger if $A > 1$.

$g(f(A))$ is bigger if $A < 1$, and $f(g(A))$ is bigger if $A > 1$.

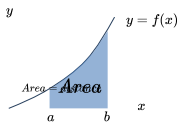
1

[MAT 2009 1A] The smallest value of

$I(a) = \int_0^1 (x^2 - a)^2\,dx$

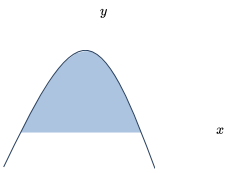
as a varies, is what?

Chapter 13 · Original slide 26 · Areas under curves



Earlier we saw that the definite integral $\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{f(x)dx}$ gives the area between a positive curve $y = f(x)$, the $\mathbf{x}$-axis, and the lines $x = a$ and $x = b$.

(We'll see why this works in a sec)



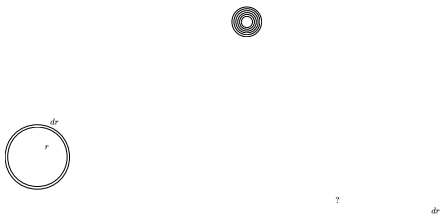
Find the area of the finite region between the curve with equation $y = 20 - x - x^2$ and the x-axis.

Factorise in order to find roots:

Therefore area between curve and x-axis is:

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 27 · Just for your interest…



Why does integrating a function give you the area under the graph?

Part 1:

You're already familiar with the idea that gradient is the rate at which a quantity changes, and we consider an infinitesimally small change in the variable.

You could consider the gradient as the little bit you're adding on each time.

Here's some practical examples using formulae you covered in your younger years!

If you wanted to consider the rate at which the area of a circle increases with respect to the radius, consider a small change dr in the radius.

The change in area is an infinitely thin ring, which looks like you've drawn a circumference.

So what is this rate?

$A = \pi r^2$
$\therefore \frac{dA}{dr} = 2\pi r$

If the added area is $2\pi r dr$ and the thickness is dr, then the length is $\frac{2\pi r dr}{dr} = 2\pi r$ as expected.

So to draw a shaded circle (which has area!), it's a bit like repeatedly drawing the circumference of a circle with gradually increasing radius.

Since the circumference is what the area is increasing by each time, the circumference is the gradient of the area of the circle, and conversely (by the definition of integration being the opposite of differentiation), the area is the integral of the circumference.

you can't add a length to an area

But by considering the infinitely thin width dr of the circumference you're drawing, it does have area!

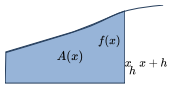
$\frac{dA}{dr} = 2\pi r \rightarrow dA = 2\pi r dr$

i.e. the change in area, dA, is $2\pi r \times dr$

If we 'roll out' the added area we're adding each time, this forms a rectangle, whose length hopefully corresponds to the circumference of the circle:

Chapter 13 · Original slide 28 · Part 2:





Consider the volume of a sphere:

$V = \frac{4}{3}\pi r^3$
$\frac{dV}{dr} = 4\pi r^2$



And the same principle applies to area under a graph. If $A(x)$ gives the area up to x, then what we're 'adding on each time' (i.e. the gradient) is sort of like drawing a line of length $f(x)$ at the right-most end each time (or more accurately, an infinitely thin rectangle of area $f(x) \times h$). Thus if $f(x)$ is the gradient of the area, then conversely the area is the integral of $f(x)$.

This works for a similar reason to before.

One final Note: Why does $\int_b^a f(x)dx$ give the area between $x = a$ and $x = b$?

This gives us an intuitive sense of why it works, but we need a formal proof:

Using the diagram above, the area up to $x + h$, i.e. $A(x + h)$, is approximately the area up to x plus the added rectangular strip:

$A(x + h) \approx A(x) + (f(x) \times h)$

Since $f(x) = A\prime(x)$, the area function $A(x)$ is the integral of $f(x)$. Thus:

$\int_b^a f(x)dx = [A(x)]_a^b = A(b) - A(a)$

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 29 · Test Your Understanding

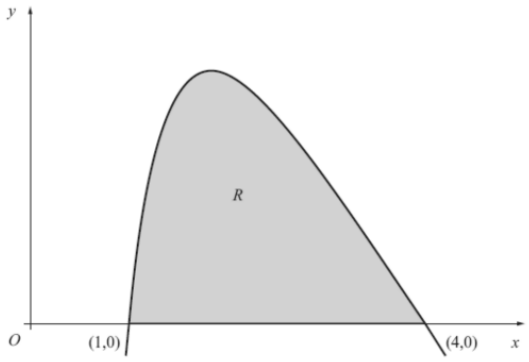


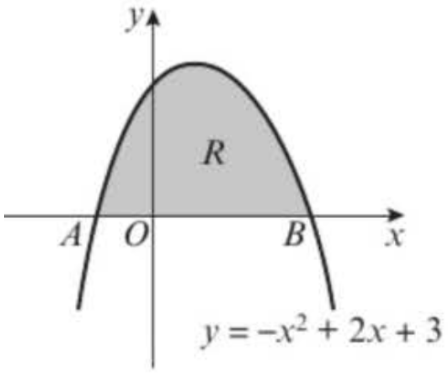
Figure 2

The finite region R, as shown in Figure 2, is bounded by the x-axis and the curve with equation $y = 27 - 2x - 9\sqrt{x} - \frac{16}{x^2}, \qquad x > 0.$

The curve crosses the x-axis at the points $(1, 0)$ and $(4, 0)$.

(c) Use integration to find the exact value for the area of R. (6)

Chapter 13 · Original slide 30 · Exercise 13E – Y1 pages 299-300



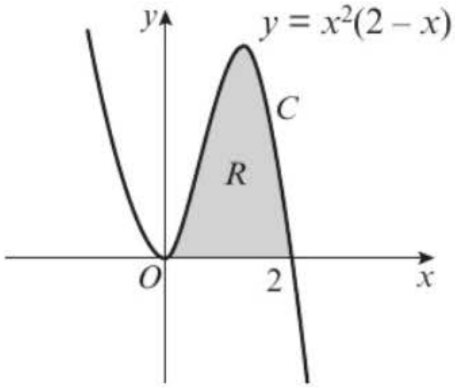
8 The finite region R is bounded by the x-axis and the curve with equation $y = -x^2 + 2x + 3$, $x \geq 0$. The curve meets the x-axis at points A and B.

a Find the coordinates of point A and point B.

b Find the area of the region R.

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 31 · Exercise 13E – Y1 pages 299-300

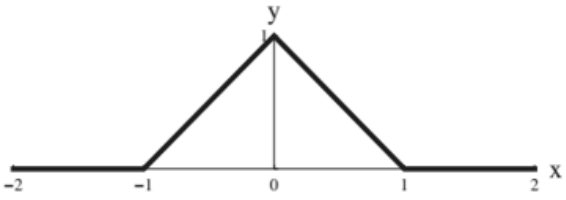


9 The graph shows part of the curve C with equation $y = x^2(2 - x)$.

The region R, shown shaded, is bounded by C and the x-axis.

Use calculus to find the exact area of R. (5 marks)

Chapter 13 · Original slide 32 · EXTENSION



1

[MAT 2007 1H] Given a function $f(x)$, you are told that

$\int_0^1 3f(x)\,dx + \int_1^2 2f(x)\,dx = 7$

$\int_0^2 f(x)\,dx + \int_1^2 f(x)\,dx = 1$

It follows that $\int_0^2 f(x)\,dx$ equals what?

2

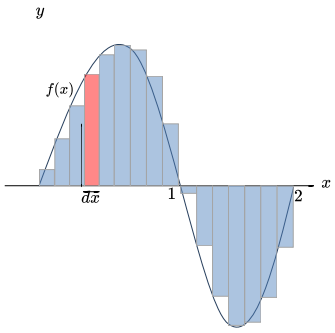
[MAT 2011 1G]

A graph of the function $y = f(x)$ is sketched on the axes below:

What is the value of $\int_{-1}^1 f(x^2 - 1)\,dx$?

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 33 · 'Negative Areas'



Sketch the curve $y = x(x-1)(x-2)$ (which expands to give $y = x^3 - 3x^2 + 2x$).

Now calculate $\int_0^2 x\,(x-1)\,(x-2)dx$. Why is this result surprising?

Integration $\int f\,(x)dx$ is just the sum of areas of infinitely thin rectangles, where the current y value (i.e. $f(x)$) is each height, and the widths are dx.

i.e. The area of each is $f\,(x) \times dx$

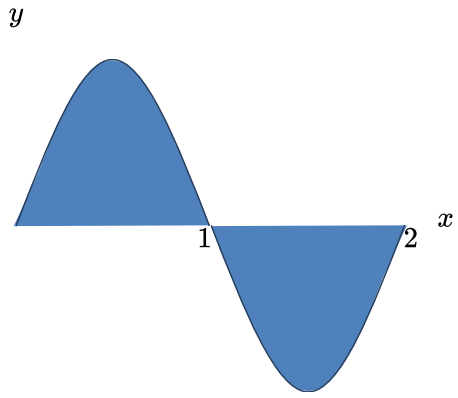
The problem is, when $f(x)$ is negative, then $f\,(x) \times dx$ is negative, i.e. a negative area!

The result is that the 'positive area' from 0 to 1 is cancelled out by the 'negative area' from 1 to 2, giving an overall 'area' of 0.

So how do we resolve this?

Note: This explains the dx in the $\int f\,(x)dx$, which effectively means "the sum of the areas of strips, each of area $f\,(x) \times dx$. So the dx is not just part of the $\int$ notation, it's behaving as a physical quantity! (i.e. length)

Chapter 13 · Original slide 34 · Example



Find the total area bound between the curve $y = x(x-1)(x-2)$ and the x-axis.

Strategy:

Separately find the area between $x = 0$ and 1, and between 1 and 2. Treat any negative areas as positive.

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 35 · Test Your Understanding

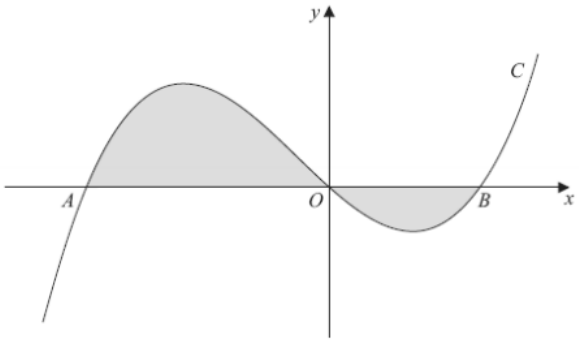


Figure 3

Figure 3 shows a sketch of part of the curve C with equation
$y = x(x + 4)(x - 2)$.

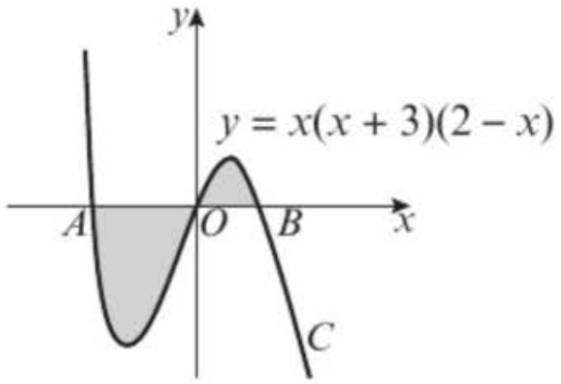
The curve C crosses the x-axis at the origin O and at the points A and B.

(a) Write down the x-coordinates of the points A and B. (1)

The finite region, shown shaded in Figure 3, is bounded by the curve C and the x-axis.

(b) Use integration to find the total area of the finite region shown shaded in Figure 3. (7)

Chapter 13 · Original slide 36 · Exercise 13F – Y1 pages 301-302



The finite region, shown shaded, is bounded by the curve C and the x-axis.

b Use integration to find the total area of the finite shaded region. (7 marks)

2 The graph shows a sketch of part of the curve C with equation $y = x(x + 3)(2 - x)$.

The curve C crosses the x-axis at the origin O and at points A and B.

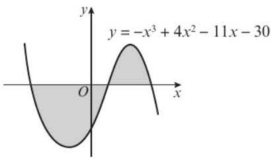
a Write down the x-coordinates of A and B. (1 mark)

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 37 · Exercise 13F – Y1 pages 301-302

$x - 30$
ketch of part of the curve with
$x^2 + 11x - 30$.
orem to show that $(x + 3)$ is a

orm $(x + 3)(Ax^2 + Bx + C)$.
(x) completely.



3 $\mathrm{f}(x) = -x^3 + 4x^2 + 11x - 30$.

The graph shows a sketch of part of the curve with equation $y = -x^3 + 4x^2 + 11x - 30$.

a Use the factor theorem to show that $(x + 3)$ is a factor of $\mathrm{f}(x)$.

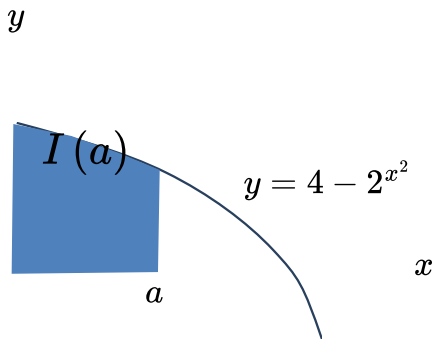
b Write $\mathrm{f}(x)$ in the form $(x + 3)(Ax^2 + Bx + C)$.

c Hence, factorise $\mathrm{f}(x)$ completely.

d Hence, determine the x-coordinates where the curve intersects the x-axis.

e Hence, determine the total shaded area shown on the sketch.

Chapter 13 · Original slide 38 · EXTENSION



Hint: It's not actually even possible to integrate 2^{x^2}, but we can still sketch the graph. Reflect on what $\frac{dI}{da}$ actually means.

Extension

1

[MAT 2010 1I] For a positive number a, let

$I(a) = \int_0^a \left(4 - 2^{x^2}\right) dx$

Then $\frac{dI}{da} = 0$ when a is what value?

$\mathbf{I(a)}$ represents the area from **x=0** up to **a**.

2

[STEP I 2014 Q3]

The numbers a and b, where $b > a \geq 0$, are such that

$\int_a^b x^2 dx = \left(\int_a^b x dx\right)^2$

(i) In the case $a = 0$ and $b > 0$, find the value of b.

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 39 · Extension

[STEP I 2014 Q3] The numbers a and b, where $b > a \geq 0$, are such that $\int_a^b x^2 dx = \left(\int_a^b x dx\right)^2$

(i) In the case $a = 0$ and $b > 0$, find the value of b.

(ii) In the case $a = 1$, show that b satisfies
$3b^3 - b^2 - 7b - 7 = 0$
Show further, with the help of a sketch, that there is only one (real) value of b that satisfies the equation and that it lies between 2 and 3.

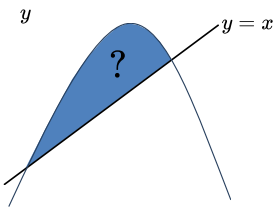
(iii) Show that $3p^2 + q^2 = 3p^2q$, where $p = b + a$ and $q = b - a$, and express p^2 in terms of q. Deduce that $1 < b - a \leq \frac{4}{3}$

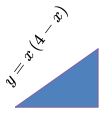
This question was actually devised to address what happens when students misunderstand or mis-apply a "rule" of mathematics and *it turns out to give the right answer*. Part (i) starts you off gently: integrating both terms, squaring the RHS and solving very quickly gives $b = \frac{4}{3}$. Part (ii) develops in much the same way, but with a non-zero lower limit to the integrals, and we immediately see that the algebra gets much more involved. Importantly, it should be very clear that whatever expression materialises must have $(b - 1)$ as a factor (since setting $b = a$ would definitely give a zero area, thus trivially satisfying the given integral statement). This leads to the required cubic equation.

The final part of (ii) requires a mixture of different ideas (and can be done in a number of different ways). The most basic approach to demonstrating that a cubic curve has only one zero is to illustrate that both of its TPs lie on the same side of the x-axis (or to show there are no TPs). The popular *Change-of-Sign Rule* for continuous functions can be used to identify the position of this zero.

Having got you started with some simple lower limits, part (iii) develops matters more generally, and derives the (perhaps) surprising result that the exploration of this initial "stupid idea" requires b and a to be "not too far apart" to an extent that is easily identifiable.

Chapter 13 · Original slide 40 · Areas between curves and lines





How could we find the area between the line and the curve?

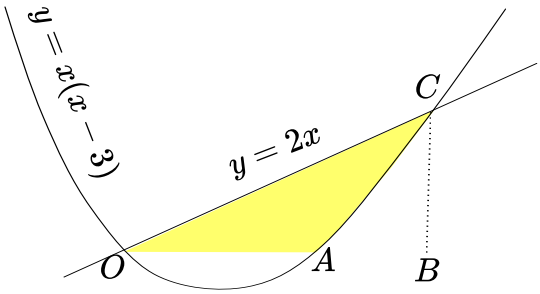
Start with the area under $\mathbf{y=x\,(4-x)}$ up to the point of intersection, then subtract the area of the triangle to 'cut it out'.

Click for animation >

Determine the area between the lines with equations $y = x\,(4 - x)$ and $y = x$

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 41 · A Harder One



[Textbook] The diagram shows a sketch of the curve with equation $y = x\,(x-3)$ and the line with equation $y = 2x$.

Find the area of the shaded region OAC.

What areas should we subtract this time?

Chapter 13 · Original slide 42 · Test Your Understanding

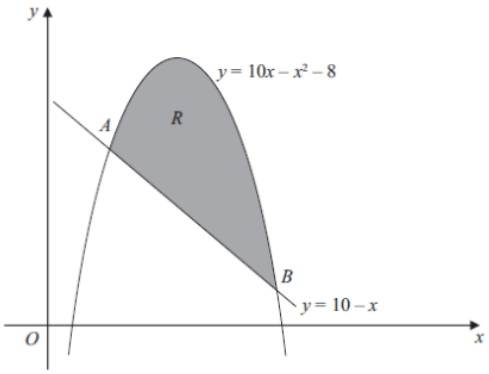


Figure 2 shows the line with equation $y = 10 - x$ and the curve with equation $y = 10x - x^2 - 8$.

The line and the curve intersect at the points A and B, and O is the origin.

(a) Calculate the coordinates of A and the coordinates of B. (5)

The shaded area R is bounded by the line and the curve, as shown in Figure 2.

(b) Calculate the exact area of R. (7)

a b

Alternative Method:

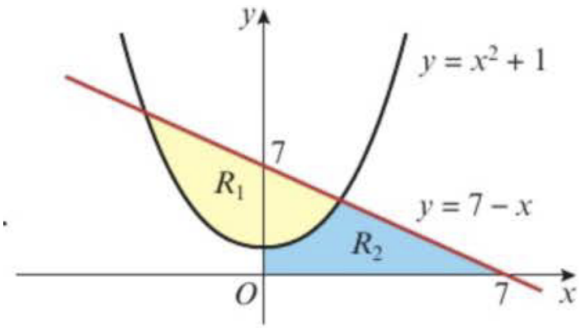
If the top curve has equation $y = f\,(x)$ and the bottom curve $y = g\,(x)$, the area between them is:

$\int_b^a (f\,(x) - g\,(x))dx$

This means you can integrate a single expression to get the final area, without any adjustment required after.

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 43 · Exercise 13G – Y1 pages 304-306



6 The diagram shows a sketch of part of the curve with equation $y = x^2 + 1$ and the line with equation $y = 7 - x$.

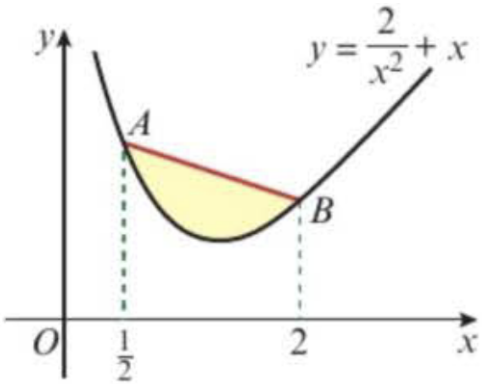
The finite region, R_1 is bounded by the line and the curve.

The finite region, R_2 is below the curve and the line and is bounded by the positive x- and y-axes as shown in the diagram.

a Find the area of R_1.

b Find the area of R_2.

Chapter 13 · Original slide 44 · Exercise 13G – Y1 pages 304-306

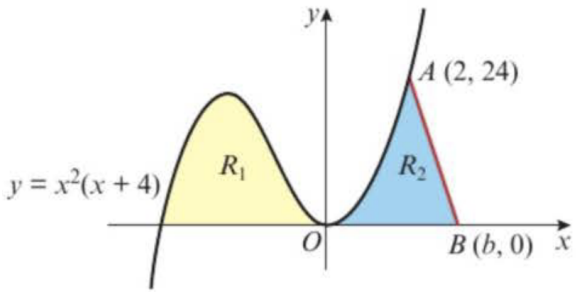


8 The diagram shows part of a sketch of the curve with equation $y = \dfrac{2}{x^2} + x$. The points A and B have x-coordinates $\frac{1}{2}$ and 2 respectively.

Find the area of the finite region between AB and the curve.

1.37 Integration: Definite Integrals and Areas

Chapter 13 · Original slide 45 · Exercise 13G – Y1 pages 304-306



10 The sketch shows part of the curve with equation $y = x^2(x + 4)$. The finite region R_1 is bounded by the curve and the negative x-axis. The finite region R_2 is bounded by the curve, the positive x-axis and AB, where $A(2, 24)$ and $B(b, 0)$.

The area of R_1 = the area of R_2.

a Find the area of R_1.

b Find the value of b.

Chapter 13 · Original slide 46 · EXTENSION

a b

[MAT 2005 1A] What is the area of the region bounded by the curves $y = x^2$ and $y = x + 2$?

[MAT 2016 1H] Consider two functions

$f(x) = a - x^2$

$g(x) = x^4 - a$

For precisely which values of $a > 0$ is the area of the region bounded by the x-axis and the curve $y = f(x)$ bigger than the area of the region bounded by the x-axis and the curve $y = g(x)$?

(Your answer should be an inequality in terms of a)